Assignment: Factoring

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1 Factoring Algorithm

Algorithm: Reduction of factoring to order-finding
Inputs: A composite number N
Outputs: A non-trivial factor of N
Runtime: $\mathcal{O}((logN)^3)$ operations. Succeeds with probability $\mathcal{O}(1)$.
Procedure:

1. If N is even, return the factor 2
2. Determine whether $N = a^b$ for integers $a \geq 1$ and $b \geq 2$, and if so return the factor a
3. Randomly choose x in the range 1 to $N-1$. If $\gcd(x, N) > 1$ then return the factor $\gcd(x, N)$.
4. Use the order-finding subroutine to find the order r of $x \mod N$.
5. If r is even and $x^{r/2} \neq -1(\mod N)$ then compute $\gcd(x^{r/2}-1, N)$ and $\gcd(x^{r/2}+1, N)$, and test to see if one of these is a non-trivial factor, returning that factor if so. Otherwise, the algorithm fails.

2 Example $N = 15$ [1]

The use of order-finding, phase estimation, and continued fraction expansions in the quantum factoring algorithm is illustrated by applying it to factor $N = 15$. First, we choose a random number which has no common factors with N; suppose we choose $x = 7$. Next, we compute the order r of x with respect to N using the quantum order-finding algorithm: begin with the state $|0\rangle|1\rangle$ and create the state

$$\frac{1}{\sqrt{2}^t}\sum_{k=0}^{2^t-1}|k\rangle|1\rangle = \frac{1}{\sqrt{2^t}}[|0\rangle + |1\rangle + |2\rangle + \ldots + |2^{t-1}\rangle]|1\rangle \tag{1}$$

by applying $t = 11$ Hadamard gates to the first register. Choosing this value of t ensures an error probability e of at most 1/4.

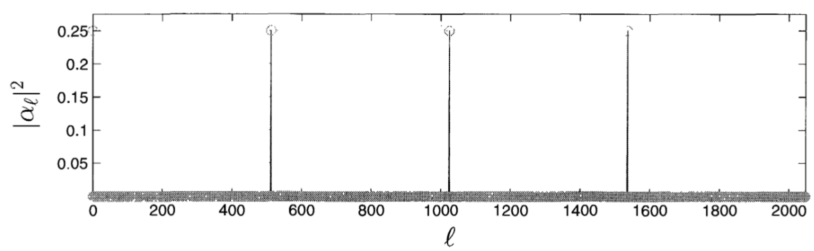
Next, compute $f(k) = x^k \mod N$, leaving the result in the second register.

$$\frac{1}{\sqrt{2}^t} \sum_{k=0}^{2^t-1} |k\rangle|x^k \mod N\rangle = |0\rangle|1\rangle+|1\rangle|7\rangle+|2\rangle|4\rangle+|3\rangle|13\rangle+|4\rangle|1\rangle+|5\rangle|7\rangle+|6\rangle|4\rangle+\ldots \tag{2}$$

We now apply the inverse Fourier transform $FT^\dagger$ to the first register and measure it. One way of analyzing the distribution of outcomes obtained is to calculate the reduced density matrix for the first register, and apply $FT^\dagger$ to it, and calculate the measurement statistics. However, since no further operation is applied to the second register, we can instead apply the principle of implicit measurement and assume that the second register is measured, obtaining a random result from $1, 7, 4$, or 13. Suppose we get 4 (any of the results works); this means the state input to $FT^\dagger$ would have been

$$\sqrt{\frac{4}{2^t}}[|2\rangle + |6\rangle + |10\rangle + |14\rangle + \ldots]. \tag{3}$$

After applying $FT^\dagger$ we obtain some state $\sum_l \alpha_l |l\rangle$, with the probability distribution



shown for $2^t = 2048$. The final measurement therefore gives either $0, 512, 1024$, or 1536, each with probability almost exactly $1/4$. Suppose we obtain $l = 1536$ from the measurement; computing the continued fraction expansion thus gives $1536/2048 = 1/(1+(1/3))$, so that $3/4$ occurs as a convergent in the expansion, giving $r = 4$ as the order of $x = 7$. By chance, r is even, and moreover, $x^{r/2} \mod N = 7^2 \mod 15 = 4 \neq -1 \mod 15$, so the algorithm works: computing the greatest common divisor $\gcd(x^2 - 1, 15) = 3$ and $\gcd(x^2 + 1, 15) = 5$ tells us that $15 = 3 \times 5$.

You will find more expansions for $\frac{1}{\sqrt{2}^t} \sum_{k=0}^{2^t-1} |k\rangle|x^k \mod N\rangle$ in [2].

3 Assignment

Apply the factoring algorithm for N and x of your choice, using the tools available in [3], where you will also find a detailed example for $x = 3$ and $N = 15$. An implicit operation to create the states after modular expponentiation is provided, but you can create this state by other means.

The assignment requires a detailed exposition of the steps to obtain the final factors, and it may include:

1. A study on the effect of the size t of the working register.
2. A comparison on the effect of different choices of x.
3. Considerations about the measurement requirements, and computational cost.

References

[1] Quantum Computation and Quantum Information, Michael A. Nielsen, Isaac L. Chuang, Cambridge University Press, 2000

[2] qiskit.org/textbook/ch-algorithms/shor.html

[3] github.com/quantummasterbarcelonacode/QuantumComputation/tree/main/Factoring